

08

Robotics & Automation
Codec Technologies India
Internship 2026

Search & Rescue Robot

LiDAR SLAM, frontier exploration
over rough terrain, camera +
TDOA victim fusion

AT A GLANCE

4/4 in 164 s

Victims found

97.0 %

Coverage of reachable area

0.34 / 1.41 m

Pose RMSE SLAM / odometry

1.4 / 22.6 deg

Yaw RMSE SLAM / odometry

0.51

Map IoU (1-cell tol.)

44.2 s

Time to first victim

60.6 m

Distance travelled

10/10

Unit tests

VERDICT

SIM PASS

TESTS 10/10

VICTIMS 4/4

DECLARED SUBSTITUTION

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1. Objective and brief mapping

The brief asks for an unknown rough-terrain map (elevation + rubble, slope-based traversability), a skid-steer robot with noisy odometry, a self-implemented 2-D LiDAR SLAM (log-odds occupancy grid + ICP or correlative scan matching) with pose error reported with and without matching, frontier exploration with A* over traversability, victim detection by HSV colour in a simulated camera and by a two-microphone cross-correlation TDOA bearing, fusion + geo-tagging, coverage tracking, a mission summary (map PNG, coverage curve, JSON, Markdown) and a GIF. All of it is implemented in **src/sar/** (numpy, scipy, OpenCV, matplotlib); the Gazebo world, slam_toolbox/Nav2 launch skeleton and ROS 2 package are provided but declared not run here.

2. Architecture

Terrain (hidden): 16 x 16 m, 0.1 m cells; smoothed-noise elevation + slab ramp; rubble blobs, 3 broken walls
traversability = $1 + 6 (\text{slope}/0.45)^2$, impassable if slope > 0.45 or occupied
SkidSteer: (v, w) -> true pose with track slip on slopes; wheel odometry 6 % scale
noise, gyro bias
0.005 rad/s + 0.05 rad/s noise; bumper/stall sensing zeroes translational odometry on a hit
Lidar2D: 121 beams / 270 deg / 6 m, 2 cm noise, 1 % dropout (ray-march on the true grid)
SLAM: OccupancyGrid log-odds (+0.85 hit, -0.5 free via cv2.fillPoly scan polygon), keyframes
ScanMatcher: likelihood field (Gaussian 0.7 cell) -> correlative search +/-0.25 m, +/-3 deg
with a motion prior; gates 0.22 m / 4 deg; ICP (KD-tree, trimmed) kept for refinement/tests
Exploration: frontiers = free cells touching unknown -> clusters -> A* (8-conn) on robot cost map
(inflated obstacles, sensed slope, bumper hazards, unknown = 3x) -> pure pursuit 0.5 m/s
after frontiers: search phase visits known-free cells the camera never looked at
Victims: camera 70 deg / 4.5 m -> HSV vest mask -> bearing + pinhole range; 2 mics 0.30 m apart,
48 kHz, 1/d amplitude -> FFT cross-correlation TDOA + parabolic peak -> bearing; tracker fuses
Mission: coverage %, pose error vs truth, GIF frames, map PNG, curves, JSON + Markdown summary

3. Implementation details

- **Mapping.** The free-space update fills the polygon spanned by the beam end points with cv2.fillPoly instead of marching each beam, which removed the radial 'unknown' stripes that appeared at long range on a 0.1 m grid and created hundreds of fake frontiers. Scans are folded into the map only from keyframes (0.15 m or 5 deg of motion) so pose jitter does not smear walls.
- **Scan matching.** I started with scan-to-map ICP and it drifted: on a grid map the thick walls give ICP a sliding degree of freedom and yaw wandered off. The final matcher is correlative (Olson 2009): brute-force (dx, dy, dtheta) over a Gaussian-blurred occupancy likelihood field, minus a motion prior around the odometry guess. The prior was the key fix - when a scan is dominated by nearby rubble it barely constrains yaw, and without the prior the estimate rotated by up to 3 deg per step in a random walk. Corrections larger than 0.22 m or 4 deg per 0.2 s step are rejected as wrong matches.
- **Rough terrain.** Steep cells are invisible to the LiDAR; the robot senses slope within 1.5 m (tilt sensor / depth camera stand-in) and re-plans when the next 1.5 m of path crosses a newly known hazard. A stall on unseen low rubble trips the bumper: translational odometry is zeroed, the cell ahead is recorded as a hazard, the robot backs up 0.24 m and re-plans. In the run there were 7 bumper hits, 21 path invalidations and 2 look-ahead stops.
- **Exploration.** Frontier clusters are scored by A* path cost minus a size bonus; goals that sit inside the inflation band are moved to the nearest finite-cost cell; failed goals are blacklisted for 15 s (not forever, because the map improves). The robot may leave an inflation band it drifted into (its own cell and 1-ring are always traversable). Once no frontier is left the planner switches to a search phase over known-free cells the camera cone has not swept.
- **Victim fusion.** A camera detection produces a geo-tag from bearing + pinhole range at the estimated pose; when the TDOA bearing agrees within 20 deg it counts double. A victim is confirmed after 3 weighted observations within 1.2 m; audio bearings without a camera hit become leads that bias the next frontier choice. Sound through rubble is attenuated 55 % (line-of-sight test).

4. Verification

All 10 tests pass (sim/test_log.txt). The end-to-end test re-runs the full mission (about 40 s of compute for 164 s of simulated time).

Group	What is checked	Tests	Result
SLAM	ICP recovers a known 7 deg / 0.29 m transform (< 0.3 deg, < 2 cm) incl. 30 % outliers; correlative match recovers a 0.14 m / 1.5 deg offset to < 5 cm / 0.6 deg; grid marks robot cell free and > 90 % of occupied cells on true obstacles	3	PASS
Planning	A* avoids inf cells, prefers the cheap corridor, returns None for a walled goal; frontiers exist after the first scan; occupied cells are inf in the cost map	2	PASS

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Victim sensing	TDOA bearing error < 2 deg over -70..70 deg and 2-6 m; audio level falls with distance; camera detects a victim at 2.5 m within 2 deg and 25 % range; tracker fuses camera + audio and geo-tags	4	PASS
Mission	seed 0: all victims found within budget, coverage >= 80 %, tolerant IoU >= 0.40, cell accuracy >= 85 %, SLAM RMSE < 0.6 m and < half of odometry, zero false geo-tags	1	PASS

5. Results

Without scan matching the map is built at the odometry pose; with the gyro bias the heading is off by 59 deg RMS, walls are drawn in the wrong place, geo-tags land 1.7 m from the true victims and only 1 of 4 victims are confirmed. The strict occupied IoU is low in both cases because a 0.3 m pose error is three cells on this grid; the 1-cell-tolerant IoU and the 90 % free/occupied accuracy are the fairer measures of a usable rescue map.

Metric	With correlative scan matching	Odometry only (control run)
Pose RMSE / max	0.34 m / 0.51 m	5.16 m / 7.84 m
Pose error at mission end	0.44 m	7.18 m
Yaw RMSE	1.4 deg	59.1 deg
Dead-reckoning error in the same run	1.41 m RMSE (odometry integrated alongside)	same estimator
Map: occupied IoU strict / 1-cell tolerance	0.18 / 0.51	0.10 / 0.31
Map: free/occupied accuracy on known cells	89.7 %	87.5 %
Coverage of reachable area / camera-swept	97.0 % / 87.1 %	99.0 % / 90.6 %
Victims found / false geo-tags	4/4 / 0	1/4 / 5
Time to first victim / all found	44.2 s / 163.4 s	38.2 s / -
Geo-tag position error (mean / max)	0.51 / 0.59 m	1.72 m
Mission time / distance / replans	164 s / 60.6 m / 61	400 s / 100.2 m / 288
Detections camera / audio / fused	88 / 769 / 84	141 / 1774 / 128
Scan matcher accepted / rejected / keyframes	811 / 6 / 540	-

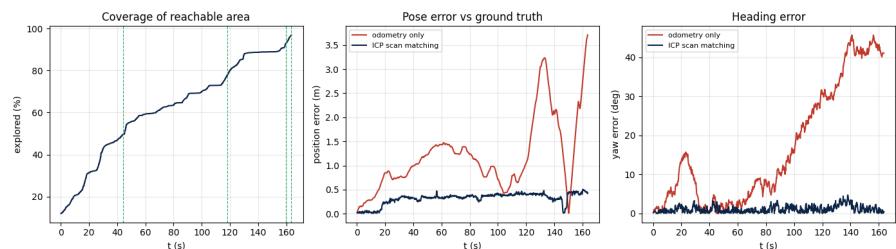


Figure 1 - curves.png: coverage of the reachable area over time (dashed = victim confirmations), position error and heading error of the scan-matched estimate versus dead reckoning.

6. Map and coverage milestones

Coverage	Reached at
25 %	16 s
50 %	45 s
75 %	116 s
90 %	156 s
95 %	162 s

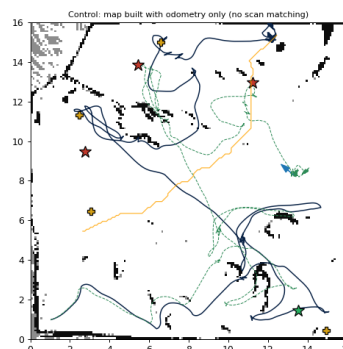


Figure 2 - map_odom_only.png (control run): the same mission mapped at the raw odometry pose. Walls are duplicated and rotated as the gyro bias accumulates; only one victim is confirmed.

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6b. Final map and exploration frame

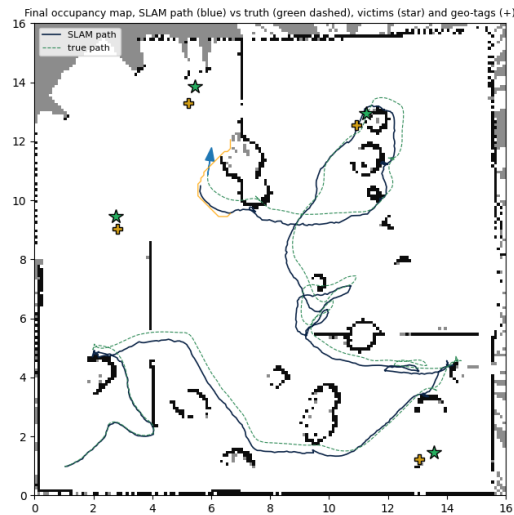


Figure 3 - map_final.png: final log-odds map (white free, black occupied, grey unknown), SLAM path (blue) vs true path (green dashed), victims (stars, green = found) and geo-tags (+).

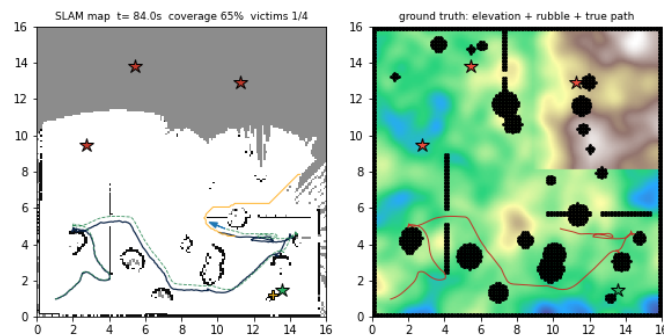


Figure 4 - a frame of exploration.gif: robot map with SLAM path, planned path (orange) and victims (left); hidden ground truth with elevation, rubble and true path (right).

7. How to run

```
cd 08_Search_Rescue_Robot
./run_all.sh # mission + odometry-only control run + GIF, then 10 unit tests;
PASS/FAIL, exit code
make sim # -> sim/summary.json, summary.md, map_final.png, curves.png,
exploration.gif, mission_log.csv
make test # python3 -m unittest discover -s tests -v
make report # docs/build_report.py -> tools/make_report.py -> docs/08_Report.pdf
cd src && python3 -m sar.mission --seed 2 --budget 300 --out ../sim # other maps
```

8. Declared substitutions / not run here

- **Gazebo:** worlds/rubble_field.sdf (16 m yard with outer walls, three broken walls with doorway gaps, rubble blocks, a slab ramp, four hi-vis victim markers and a skid-steer robot with gpu_lidar 121 x 270 deg / 6 m, a 160 x 120 camera, IMU and DiffDrive plugin) is well-formed XML but was NOT run - Gazebo is not installed here.
- **slam_toolbox / Nav2:** ros2_ws/src/search_rescue_robot/config/slam_toolbox_params.yaml and nav2_params.yaml plus the launch file's use_external_slam:=true branch are provided as a skeleton and were NOT run.
- **ROS 2 Humble:** the sar_node (subscribes /scan, /odom, /camera/image_raw, /audio/stereo; publishes /map, /sar/pose, /cmd_vel, /sar/victims, /sar/status) compiles with py_compile but was NOT launched - no rcipy in the container. Build with colcon build --packages-select search_rescue_robot; ros2 launch search_rescue_robot search_rescue_robot.launch.py.
- Everything else - terrain, robot, LiDAR, SLAM, exploration, camera/audio models, fusion, mission, GIF, tests - ran in this container with Python 3.11; all numbers above are read from sim/summary.json, sim/mission_log.csv and sim/test_log.txt by docs/build_report.py.

9. Learnings / next steps

Most of my time went into making the SLAM stable rather than writing it: ICP that looked perfect on synthetic point clouds slid along grid walls, a matcher without a motion prior let yaw random-walk when only nearby rubble was visible, and stall events fed metres of phantom odometry into the filter until I modelled a bumper. Each failure showed up first as exploration misbehaviour (fake frontiers, sealed-in robot), so the mission-level metrics were the best debugging tool. Next steps: multi-resolution correlative search for larger windows, loop closure over keyframes (pose graph) to remove the residual 0.3-0.4 m offset visible between the SLAM and true paths, a proper robot footprint in the collision model, and a run through slam_toolbox + Nav2 in Gazebo to compare against my own stack.

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